CBSE Class 12 Physics — Alternating Current

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (Objective/VSA)

Q1. If the frequency of an AC source is made 4 times its initial value, the inductive reactance of an inductor in the circuit will become: (a) 2 times (b) 3 times (c) 4 times (d) Unchanged [PYQ 2022] | **[Confidence: Very High]**

Q2. When an alternating voltage $E = E_0 \sin(\omega t)$ is applied to a circuit, a current $I = I_0 \sin(\omega t + \pi/2)$ flows through it. The average power dissipated in the circuit is: (a) $E_{rms}I_{rms}$ (b) $E_0 I_0$ (c) $E_0 I_0/\sqrt{2}$ (d) Zero [PYQ 2021] | **[Confidence: Very High]**

Q3. The voltages across a resistor, an inductor, and a capacitor connected in series to an AC source are $20\ \mathrm{V}$, $15\ \mathrm{V}$, and $30\ \mathrm{V}$ respectively. The resultant voltage of the circuit is: (a) $5\ \mathrm{V}$ (b) $20\ \mathrm{V}$ (c) $25\ \mathrm{V}$ (d) $65\ \mathrm{V}$ [PYQ 2021] | **[Confidence: Very High]**

Q4. A $20\ \mathrm{V}$ AC is applied to a circuit consisting of a resistance and a coil with negligible resistance. If the voltage across the resistance is $12\ \mathrm{V}$, the voltage across the coil is: (a) $16\ \mathrm{V}$ (b) $10\ \mathrm{V}$ (c) $8\ \mathrm{V}$ (d) $6\ \mathrm{V}$ [PYQ 2021] | **[Confidence: High]**

Q5. In a circuit, the phase difference between the alternating current and the source voltage is exactly $\pi/2$. Which of the following CANNOT be the sole element(s) of the circuit? (a) Only C (b) Only L (c) L and R (d) L or C [PYQ 2021] | **[Confidence: High]**

Q6. The power factor of a series LCR circuit at resonance is: (a) 0.707 (b) 1 (c) Zero (d) 0.5 [PYQ 2020] | **[Confidence: Very High]**

Q7. The instantaneous values of emf and the current in a series AC circuit are $E = E_0 \sin(\omega t)$ and $I = I_0 \sin(\omega t + \pi/3)$ respectively. The circuit is: (a) necessarily an RL circuit (b) necessarily an RC circuit (c) necessarily an LCR circuit (d) can be an RC or LCR circuit [PYQ 2021] | **[Confidence: High]**

Q8. A circuit is connected to an AC source of variable frequency. As the frequency of the source is increased, the current first increases and then decreases. Which of the following combinations of elements is likely to comprise the circuit? (a) L, C, and R (b) L and C (c) L and R (d) R and C [PYQ 2021] | **[Confidence: Very High]**

Q9. The rms current in a circuit connected to a $50\ \mathrm{Hz}$ AC source is $15\ \mathrm{A}$. The value of the current in the circuit exactly $\frac{1}{600}\ \mathrm{s}$ after the instant the current is zero, is: (a) $15/\sqrt{2}\ \mathrm{A}$ (b) $15\sqrt{2}\ \mathrm{A}$ (c) $15\sqrt{3/2}\ \mathrm{A}$ (d) $15\sqrt{6}\ \mathrm{A}$ [PYQ 2021] | **[Confidence: Medium]**

Q10. The impedance (Z) of a series LCR circuit is given by: (a) $R + X_L + X_C$ (b) $\sqrt{(1/X_C)^2 + (1/X_L)^2 + R^2}$ (c) $\sqrt{X_L^2 - X_C^2 + R^2}$ (d) $\sqrt{R^2 + (X_L - X_C)^2}$ [PYQ 2021] | **[Confidence: Very High]**

Q11. What is the average value of an AC voltage $V = V_0 \sin(\omega t)$ over a complete cycle? (a) $V_0/\sqrt{2}$ (b) $2V_0/\pi$ (c) V_0/π (d) Zero [Practice] | **[Confidence: Very High]**

Q12. What is the average value of the same AC voltage $V = V_0 \sin(\omega t)$ over just the positive half cycle? (a) $V_0/\sqrt{2}$ (b) $2V_0/\pi$ (c) $V_0/2$ (d) Zero [Practice] | **[Confidence: High]**

Q13. Calculate the capacitive reactance of a $50\ \mathrm{pF}$ capacitor connected to an AC source of frequency $1\ \mathrm{kHz}$. (a) $\frac{10^7}{\pi}\ \Omega$ (b) $\frac{10^6}{\pi}\ \Omega$ (c) $2\pi \times 10^7\ \Omega$ (d) $\pi \times 10^6\ \Omega$ [Practice] | **[Confidence: Medium]**

Q14. The selectivity (quality factor Q) of a series LCR AC circuit is large when: (a) L is large and R is large (b) L is small and R is small (c) L is large and R is small (d) $L = R$ [PYQ 2020] | **[Confidence: High]**

Q15. Which of the following graphs represents the correct variation of capacitive reactance (X_C) with frequency (ν)? (a) A straight line through the origin (b) A rectangular hyperbola (c) A parabola opening upwards (d) A straight line parallel to the frequency axis [PYQ 2022] | **[Confidence: Very High]**

Q16. A $300\ \Omega$ resistor and a capacitor of $\frac{25}{\pi}\ \mu\mathrm{F}$ are connected in series to a $200\ \mathrm{V} - 50\ \mathrm{Hz}$ AC source. The impedance of the circuit is: (a) $300\ \Omega$ (b) $400\ \Omega$ (c) $500\ \Omega$ (d) $700\ \Omega$ [PYQ 2021] | **[Confidence: High]**

Q17. The core of a transformer is laminated to specifically reduce energy losses due to: (a) Flux leakage (b) Copper loss (c) Hysteresis (d) Eddy currents [Practice] | **[Confidence: Very High]**

Q18. Assertion (A): A step-up transformer cannot be used as a step-down transformer. **Reason (R):** A transformer works only in one direction. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) Both A and R are false. [PYQ 2021] | **[Confidence: High]**

Q19. Assertion (A): In a purely inductive circuit, current lags behind the voltage by $\pi/2$. **Reason (R):** Inductive reactance is directly proportional to the frequency of the applied AC source. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) Both A and R are false. [Practice] | **[Confidence: Very High]**

Q20. Assertion (A): At resonance in a series LCR circuit, the impedance of the circuit is purely resistive. **Reason (R):** At resonance, the inductive reactance precisely cancels out the capacitive reactance. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) Both A and R are false. [Practice] | **[Confidence: Very High]**

2-Mark Questions

Q21. The power factor of an AC circuit is 0.5. What is the exact phase difference between the voltage and the current in the circuit? [PYQ 2016] | **[Confidence: Very High]**

Q22. A $100\ \Omega$ resistor is connected to a $220\ \mathrm{V}, 50\ \mathrm{Hz}$ AC supply. Calculate: (A) the peak value of the current in the circuit, and (B) the net power consumed over a full cycle. [NCERT/PYQ] | **[Confidence: Very High]**

Q23. Why is the use of AC voltage vastly preferred over DC voltage for long-distance transmission of electrical energy? Give two explicit reasons. [PYQ 2014] | **[Confidence: Very High]**

Q24. Why does current not flow continuously in a capacitor connected across a steady DC battery? However, a momentary current does flow during the charging or discharging of the capacitor. Explain conceptually. [PYQ 2017] | **[Confidence: High]**

Q25. An inductor L of reactance X_L is connected in series with a glowing bulb B to an AC source. Briefly explain how the brightness of the bulb changes when a thick iron rod is fully inserted inside the inductor. [PYQ 2016] | **[Confidence: Very High]**

Q26. A step-down transformer is used to reduce the main supply of $220\ \mathrm{V}$ to $22\ \mathrm{V}$. If the currents in the primary and secondary are $2\ \mathrm{A}$ and $15\ \mathrm{A}$ respectively, calculate the efficiency of the transformer. [PYQ 2022] | **[Confidence: High]**

Q27. Define the term 'Wattless Current'. State the specific condition under which a wattless current flows in an AC circuit. [PYQ 2014] | **[Confidence: High]**

Q28. A resistor R and an inductor L are connected in series to a source of voltage $V = V_0 \sin(\omega t)$. The voltage is found to lead the current in phase by $\pi/4$. If the inductor is replaced by a capacitor C, the voltage lags behind the current by $\pi/4$. When L, C, and R are all connected in series with the same source, what is the exact average power dissipated? [PYQ 2020] | **[Confidence: High]**

Q29. In a series LCR circuit, write the mathematical conditions under which: (A) The impedance of the circuit is strictly minimum. (B) What is this specific state of the circuit called? [PYQ 2014] | **[Confidence: Very High]**

Q30. Plot a neat graph showing the variation of impedance (Z) of a series LCR circuit with the angular frequency (ω) of the AC source. Clearly mark the resonant frequency (ω_r) on the graph. [PYQ 2020] | **[Confidence: High]**

Q31. The primary coil of an ideal transformer has 100 turns, and the secondary has 200 turns. A $220\ \mathrm{V}$ input at $10\ \mathrm{A}$ is supplied to the primary. Find the output voltage and output current. [PYQ 2022] | **[Confidence: Medium]**

Q32. Define 'Capacitive Reactance'. Write its SI unit and state how it varies with the frequency of the AC source. [PYQ 2015] | **[Confidence: Very High]**

3-Mark Questions

Q33. An AC voltage $V = V_0 \sin(\omega t)$ is applied across a pure inductor of self-inductance L. Find the mathematical expression for the current flowing in the circuit and show that the current strictly lags behind the applied voltage by a phase angle of $\pi/2$. [PYQ 2022] | **[Confidence: Very High]**

Q34. An AC source generating a voltage $V = V_0 \sin(\omega t)$ is connected to a pure capacitor of capacitance C. Derive the expression for the current I flowing through it. Plot a graph of V and I versus ωt to show that the current is ahead of the voltage by $\pi/2$. [PYQ 2022/2014] | **[Confidence: Very High]**

Q35. A source of AC voltage $V = V_m \sin(\omega t)$ is connected to a series combination of a capacitor C and a resistor R. Draw the corresponding phasor diagram and use it to rigorously obtain the expressions for: (A) the impedance of the circuit, and (B) the phase angle. [PYQ 2015] | **[Confidence: High]**

Q36. A capacitor of unknown capacitance, a resistor of $100\ \Omega$, and an inductor of self-inductance $L = 4/\pi^2\ \mathrm{H}$ are connected in series to an AC source of $200\ \mathrm{V}$ and $50\ \mathrm{Hz}$. Calculate the value of the capacitance and the total impedance of the circuit when the current is exactly in phase with the voltage. Calculate the power dissipated in the circuit. [PYQ 2016] | **[Confidence: Very High]**

Q37. The figure shows a series LCR circuit connected to a variable frequency $250\ \mathrm{V}$ source with $L = 50\ \mathrm{mH}$, $C = 80\ \mu\mathrm{F}$, and $R = 40\ \Omega$. Determine: (A) The source frequency which drives the circuit into resonance, and (B) The quality factor (Q) of the circuit. [PYQ 2014] | **[Confidence: High]**

Q38. Show mathematically that an ideal pure inductor does not dissipate any average power in an AC circuit over a complete cycle. [PYQ 2020] | **[Confidence: High]**

Q39. A power transmission line feeds input power at $2200\ \mathrm{V}$ to a step-down transformer with its primary windings having 3000 turns. Find the required number of turns in the secondary winding to get the power output at $220\ \mathrm{V}$. [PYQ 2017] | **[Confidence: Medium]**

Q40. Explain with an example how severe power loss is reduced if electrical energy is transmitted over long distances as an alternating current at a very high voltage rather than a direct current at a low voltage. [PYQ 2018] | **[Confidence: Very High]**

Q41. A device 'X' is connected to an AC source $V = V_0 \sin(\omega t)$. The current through X is given by $I = I_0 \sin(\omega t + \pi/2)$. (A) Identify the device X and write the mathematical expression for its reactance. (B) How does the reactance of the device X vary with the frequency of the AC? Show this graphically. [PYQ 2018] | **[Confidence: Very High]**

Q42. In a series LCR circuit connected across an AC source of variable frequency, what is the exact phase difference between the voltages across the inductor (V_L) and the capacitor (V_C) at resonance? Justify by drawing a brief phasor diagram. [PYQ 2019] | **[Confidence: High]**

5-Mark Questions (Long Answer/Case-Study)

Q43. Case Study: Transformer and its Working For many purposes, it is absolutely necessary to change (or transform) an alternating voltage from one to another of greater or smaller value. This is done with a device called a transformer, which operates on the principle of mutual induction. When an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf in it. We consider an ideal transformer in which the primary has negligible resistance and all the flux in the core uniquely links both primary and secondary windings. (A) What is the core difference in the number of turns between a step-up and a step-down transformer? (1 Mark) (B) A transformer is used to reduce the mains supply of $220\ \mathrm{V}$ to $22\ \mathrm{V}$. If the currents in the primary and secondary are $2\ \mathrm{A}$ and $15\ \mathrm{A}$ respectively, what is the efficiency of the transformer? (1.5 Marks) (C) Can a transformer be used to step up a DC voltage? State a scientific reason for your answer. (1.5 Marks) (D) Name any two specific causes of energy loss in an actual transformer. (1 Mark) [PYQ 2022] | **[Confidence: Very High]**

Q44. (A) A series LCR circuit is connected to an AC source having voltage $V = V_m \sin(\omega t)$. Using a phasor diagram, derive the expression for the instantaneous current I and its phase relationship (ϕ) to the applied voltage. (B) Obtain the condition for resonance to occur. (C) Define 'Power Factor'. State the exact conditions under which it is (i) maximum, and (ii) minimum. [PYQ 2019] | **[Confidence: Very High]**

Q45. (A) Draw a schematic, labelled diagram of an AC generator. State its underlying working principle. (B) Obtain the mathematical expression for the instantaneous value of the emf (ε) induced in the rotating coil of N turns, each of cross-sectional area A, in the presence of a uniform magnetic field B, rotating with an angular frequency ω. (C) Plot a graph showing the variation of the alternating emf generated by a loop of wire rotating in a magnetic field with respect to time over one complete cycle. [PYQ 2019/2017] | **[Confidence: Very High]**

Q46. (A) Explain with the help of a labelled diagram, the principle and working of a transformer. (B) Deduce the expression relating the ratio of output voltage to input voltage with the number of turns in the primary and secondary coils. (C) The primary coil of an ideal step-up transformer has 100 turns and the transformation ratio is 100. The input voltage and input power are $220\ \mathrm{V}$ and $1100\ \mathrm{W}$ respectively. Calculate: (i) the number of turns in the secondary, (ii) the current in the primary, and (iii) the voltage across the secondary. [PYQ 2016] | **[Confidence: Very High]**

Q47. (A) In a series LCR circuit connected across an AC source of variable frequency, obtain the expression for its impedance Z. (B) A $2\ \mu\mathrm{F}$ capacitor, $100\ \Omega$ resistor, and an $8\ \mathrm{H}$ inductor are connected in series with an AC source. (i) What should be the frequency of the source such that the current drawn in the circuit is maximum? What is this specific frequency called? (ii) If the peak value of the emf of the source is $200\ \mathrm{V}$, find the maximum current in the circuit. [PYQ 2016] | **[Confidence: High]**

Q48. A device 'X' is connected to an AC source $V = V_m \sin(\omega t)$. The variation of voltage, current, and power in one complete cycle is plotted, showing that the current lags the voltage by exactly $\pi/2$. (A) Identify the device 'X' (assuming it's a pure element). (B) Derive the expression for the current in the circuit and its phase relation with the AC voltage. (C) How does its impedance vary with the frequency of the AC source? Show this graphically. (D) When this device is connected to a $200\ \mathrm{V}$

DC voltage, a current of $1\ \mathrm{A}$ flows through it. When the same device is connected to a $200\ \mathrm{V}, 50\ \mathrm{Hz}$ AC source, only $0.5\ \mathrm{A}$ current flows. Explain why this happens, and calculate the self-inductance of the device. [PYQ 2017/2019] | **[Confidence: Very High]**

Q49. (A) State the condition for electrical resonance in a series LCR circuit. (B) Plot a graph showing the variation of current with the frequency of the applied voltage for two different values of resistance R_1 and R_2 ($R_1 > R_2$). (C) A sinusoidal voltage of peak value $283\ \mathrm{V}$ and frequency $50\ \mathrm{Hz}$ is applied to a series LCR circuit in which $R = 3\ \Omega$, $L = 25.48\ \mathrm{mH}$, and $C = 796\ \mu\mathrm{F}$. Find: (i) the impedance of the circuit, (ii) the phase difference between the voltage across the source and the current, and (iii) the power dissipated in the circuit. [Practice/NCERT] | **[Confidence: High]**

Q50. (A) Explain the term 'wattless current' in an AC circuit. (B) A $15\ \Omega$ resistor, an $80\ \mathrm{mH}$ inductor, and a capacitor of capacitance C are connected in series with a $50\ \mathrm{Hz}$ AC source. If the source voltage and current in the circuit are exactly in phase, calculate the value of capacitance C. (C) What is the average power dissipated by the capacitor and the inductor in this circuit over a complete cycle? Justify. [PYQ 2021 / Practice] | **[Confidence: High]**

Answer Key

Q1: (c) 4 times ($X_L = \omega L = 2\pi f L$. If f is 4 times, X_L is 4 times). **Q2:** (d) Zero (Phase difference is $\pi/2$. $P = E_{rms} I_{rms} \cos(\pi/2) = 0$). **Q3:** (c) $25\ \mathrm{V}$ ($V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{20^2 + (15-30)^2} = \sqrt{400+225} = \sqrt{625} = 25\ \mathrm{V}$). **Q4:** (a) $16\ \mathrm{V}$ ($V^2 = V_R^2 + V_L^2 \Rightarrow 20^2 = 12^2 + V_L^2 \Rightarrow 400 = 144 + V_L^2 \Rightarrow V_L = \sqrt{256} = 16\ \mathrm{V}$). **Q5:** (c) L and R (An LR circuit has a phase difference strictly between 0 and $\pi/2$, never exactly $\pi/2$). **Q6:** (b) 1 (At resonance, $Z = R$, so $\cos\phi = R/Z = R/R = 1$). **Q7:** (d) can be an RC or LCR circuit (Current leads voltage by $\pi/3$, meaning the circuit is capacitive in nature. Pure RC or LCR with $X_C > X_L$ works). **Q8:** (a) L, C, and R (This is the defining characteristic of a resonance curve in an LCR series circuit). **Q9:** (c) $15\sqrt{3/2}\ \mathrm{A}$ ($I_0 = 15\sqrt{2}$. $I = I_0 \sin(\omega t) = 15\sqrt{2}\sin(100\pi \times \frac{1}{600}) = 15\sqrt{2}\sin(30^\circ) = 15\sqrt{2} \times 0.5 = 15/\sqrt{2} = 15\sqrt{2}/2 = 15\sqrt{2/4} = 15\sqrt{1/2}$. Wait, $\sin(100\pi/600) = \sin(\pi/6) = 1/2$. $I = 15\sqrt{2}/2 = 15\sqrt{2/4} = 15/\sqrt{2} = 15\sqrt{2}/2 = 10.6\ \mathrm{A}$. Actually, option (a) $15/\sqrt{2}$ matches this. My answer key uses $15/\sqrt{2}$). *Correction: Answer is (a).* **Q10:** (d) $\sqrt{R^2 + (X_L - X_C)^2}$. **Q11:** (d) Zero (The positive and negative half cycles cancel each other out). **Q12:** (b) $2V_0/\pi$. **Q13:** (a) $\frac{10^7}{\pi}\ \Omega$ ($X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 10^3 \times 50 \times 10^{-12}} = \frac{1}{100\pi \times 10^{-9}} = \frac{10^7}{\pi}\ \Omega$). **Q14:** (c) L is large and R is small ($Q = \frac{1}{R}\sqrt{\frac{L}{C}}$). **Q15:** (b) A rectangular hyperbola ($X_C \propto 1/\nu$). **Q16:** (c) $500\ \Omega$ ($X_C = \frac{1}{2\pi \times 50 \times (25/\pi) \times 10^{-6}} = \frac{1}{2500 \times 10^{-6}} = 400\ \Omega$. $Z = \sqrt{300^2 + 400^2} = 500\ \Omega$). **Q17:** (d) Eddy currents (Laminations heavily increase resistance to circulating eddy currents). **Q18:** (d) Both A and R are false (A transformer can be used in reverse, working in both directions, provided voltage ratings are not dangerously exceeded). **Q19:** (b) Both A and R are true but R is NOT the correct explanation of A. **Q20:** (a) Both A and R are true and R is the correct explanation of A. **Q21:** $\cos\phi = 0.5 \Rightarrow \phi = \cos^{-1}(0.5) = 60^\circ$ (or $\pi/3$ radians). **Q22:** (A) $I_{rms} = V_{rms}/R = 220/100 = 2.2\ \mathrm{A}$. Peak current $I_0 = I_{rms}\sqrt{2} = 2.2\sqrt{2} \approx 3.11\ \mathrm{A}$. (B) $P = V_{rms} I_{rms} \cos(0^\circ) = 220 \times 2.2 = 484\ \mathrm{W}$. **Q23:** (1) AC can be easily stepped up to high voltages using transformers, heavily reducing I^2R power losses during transmission. (2) AC can be easily converted to DC when needed. **Q24:** In DC, frequency $f = 0$, so $X_C = 1/(2\pi f C) = \infty$ (acts as an open circuit). During charging/discharging, voltage is changing ($dV/dt \neq 0$), producing a displacement current. **Q25:** Brightness tightly decreases. Inserting the iron rod increases the magnetic permeability, increasing L and X_L. Impedance Z increases, so the current I drawn drops, dimming the bulb. **Q26:** $\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{V_s I_s}{V_p I_p} \times 100 = \frac{22 \times 15}{220 \times 2} \times 100 = \frac{330}{440} \times 100 = 75\%$. **Q27:** Current in an AC circuit which consumes exactly zero average power. Condition: Phase difference between V and I must be exactly $\pi/2$ (occurs in purely inductive or purely capacitive circuits). **Q28:** When L is present, $\tan(\pi/4) = X_L/R \Rightarrow X_L = R$. When C is present, $\tan(\pi/4) = X_C/R \Rightarrow X_C = R$. In LCR, $X_L = X_C$, so it's in resonance. $Z = R$. Average power $P = V_{rms}^2/R$. **Q29:** (A) Minimum impedance happens when $X_L = X_C$ or $\omega L = 1/(\omega C)$. (B) Electrical Resonance. **Q30:** Graph is a U-shaped/V-shaped curve. Z is minimum ($Z = R$) at $\omega = \omega_r$, and increases on both sides. **Q31:** $V_s = V_p(N_s/N_p) = 220 \times (200/100) = 440\ \mathrm{V}$. $I_s = I_p(N_p/N_s) = 10 \times (100/200) = 5\ \mathrm{A}$. **Q32:** The effective opposition offered by a capacitor to the flow of AC. SI unit is Ohm (Ω). $X_C = 1/(2\pi f C)$, it varies inversely with frequency. **Q33:** $V - L(dI/dt) = 0 \Rightarrow dI = (V_0/L)\sin(\omega t)dt$. Integrating gives $I = -(V_0/\omega L)\cos(\omega t) = I_0 \sin(\omega t - \pi/2)$. **Q34:** $Q = CV \Rightarrow I = dQ/dt = C(dV/dt) = C(V_0 \omega \cos(\omega t)) = (V_0/X_C)\sin(\omega t + \pi/2)$. Graph shows I peaking a quarter cycle before V. **Q35:** Draw V_R along the x-axis (with I), and V_C pointing vertically downwards. $V = \sqrt{V_R^2 + V_C^2}$. $Z = \sqrt{R^2 + X_C^2}$. $\tan\phi = X_C/R$. **Q36:** I and V in phase means resonance: $\omega L = 1/(\omega C) \Rightarrow C = 1/(\omega^2 L) = 1/((100\pi)^2 \times 4/\pi^2) = 1/40000 = 25\ \mu\mathrm{F}$. Impedance $Z = R = 100\ \Omega$.

$P = V_{rms}^2/R = 200^2/100 = 400\text{ W}$. **Q37:** (A) $f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{50\times10^{-3}\times80\times10^{-6}}} = \frac{1}{2\pi\sqrt{4\times10^{-6}}} = \frac{1000}{4\pi} \approx 79.6\text{ Hz}$. (B) $Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{40}\sqrt{\frac{50\times10^{-3}}{80\times10^{-6}}} = \frac{1}{40}\sqrt{625} = \frac{25}{40} = 0.625$. **Q38:** $P_{avg} = V_{rms}I_{rms}\cos(\pi/2)$. Since $\cos(\pi/2) = 0$, $P_{avg} = 0$. **Q39:** $N_s = N_p(V_s/V_p) = 3000 \times (220/2200) = 3000 \times 0.1 = 300$ turns. **Q40:** High voltage implies low current ($P = VI$). Power lost in wires is I^2R. Since I is heavily reduced, the I^2R heating loss is exponentially reduced. **Q41:** (A) Device X is a pure capacitor (current leads by $\pi/2$). Reactance $X_C = 1/(\omega C)$. (B) Reactance varies inversely with frequency (rectangular hyperbola). **Q42:** Exactly 180° (or π radians). Phasor diagram shows V_L pointing $+90^\circ$ (up) from I, and V_C pointing -90° (down) from I. **Q43:** (A) Step-up has $N_s > N_p$; step-down has $N_s < N_p$. (B) $\eta = (22 \times 15)/(220 \times 2) = 330/440 = 75\%$. (C) No, DC creates a steady magnetic flux. Mutual induction heavily requires a *changing* magnetic flux. (D) Copper loss (I^2R heating), Eddy currents, Hysteresis loss, Flux leakage. **Q44:** (A) Draw V_R, V_L, V_C phasors. Resultant $V = \sqrt{V_R^2 + (V_L - V_C)^2}$. Current $I = V/Z$. $\tan\phi = (X_L - X_C)/R$. (B) Resonance: $X_L = X_C$. (C) Power factor $\cos\phi = R/Z$. (i) Max (1) at resonance ($Z = R$). (ii) Min (0) in pure L or pure C circuits. **Q45:** (A) Diagram: Armature, field magnets, slip rings, brushes. Principle: Electromagnetic Induction. (B) $\phi = NBA\cos(\omega t) \Rightarrow \varepsilon = -d\phi/dt = NBA\omega\sin(\omega t)$. (C) Sine wave graph from 0 to $2\pi/\omega$. **Q46:** (A) Diagram of core, primary and secondary coils. Working via mutual induction. (B) $\varepsilon_p = -N_p(d\phi/dt)$, $\varepsilon_s = -N_s(d\phi/dt) \Rightarrow V_s/V_p = N_s/N_p$. (C) (i) $N_s = 100 \times 100 = 10,000$. (ii) $I_p = P/V_p = 1100/220 = 5\text{ A}$. (iii) $V_s = 220 \times 100 = 22,000\text{ V}$. **Q47:** (A) $Z = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$. (B) (i) Current is max at resonant frequency $f_r = 1/(2\pi\sqrt{LC}) = 1/(2\pi\sqrt{8 \times 2 \times 10^{-6}}) = 1/(2\pi \times 4 \times 10^{-3}) = 1000/(8\pi) \approx 39.8\text{ Hz}$. (ii) At resonance, $Z = R = 100\ \Omega$. $I_0 = V_0/R = 200/100 = 2\text{ A}$. **Q48:** (A) Pure Inductor. (B) $V = L(dI/dt) \Rightarrow I = (V_m/\omega L)\sin(\omega t - \pi/2)$. (C) $X_L = \omega L$, straight line through origin. (D) In DC, inductor acts as a pure resistor: $R = 200/1 = 200\ \Omega$. In AC, $Z = 200/0.5 = 400\ \Omega$. $Z^2 = R^2 + X_L^2 \Rightarrow 400^2 = 200^2 + X_L^2 \Rightarrow X_L = \sqrt{160000 - 40000} = \sqrt{120000} = 200\sqrt{3}\ \Omega$. $L = X_L/2\pi f = 200\sqrt{3}/(100\pi) = 2\sqrt{3}/\pi \approx 1.1\text{ H}$. **Q49:** (A) $X_L = X_C$. (B) Resonance curves: sharper, taller peak for R_2 (since $R_1 > R_2$). (C) (i) $X_L = 2\pi \times 50 \times 25.48 \times 10^{-3} \approx 8\ \Omega$. $X_C = 1/(2\pi \times 50 \times 796 \times 10^{-6}) \approx 4\ \Omega$. $Z = \sqrt{3^2 + (8-4)^2} = \sqrt{9+16} = 5\ \Omega$. (ii) $\tan\phi = (8-4)/3 = 4/3 \Rightarrow \phi = 53.1^\circ$. (iii) $V_{rms} = 283/\sqrt{2} = 200\text{ V}$. $I_{rms} = 200/5 = 40\text{ A}$. $P = I_{rms}^2 R = 40^2 \times 3 = 4800\text{ W}$. **Q50:** (A) Wattless current is the component of AC ($I_{rms}\sin\phi$) that does zero work/dissipates no power, arising when phase angle is 90°. (B) In phase means resonance: $X_L = X_C \Rightarrow C = 1/((100\pi)^2 \times 80 \times 10^{-3}) = 1/(10000\pi^2 \times 0.08) \approx 126.6\ \mu\text{F}$. (C) Exactly zero. Pure inductors and capacitors store energy in one quarter cycle and return it in the next, causing zero net power dissipation.